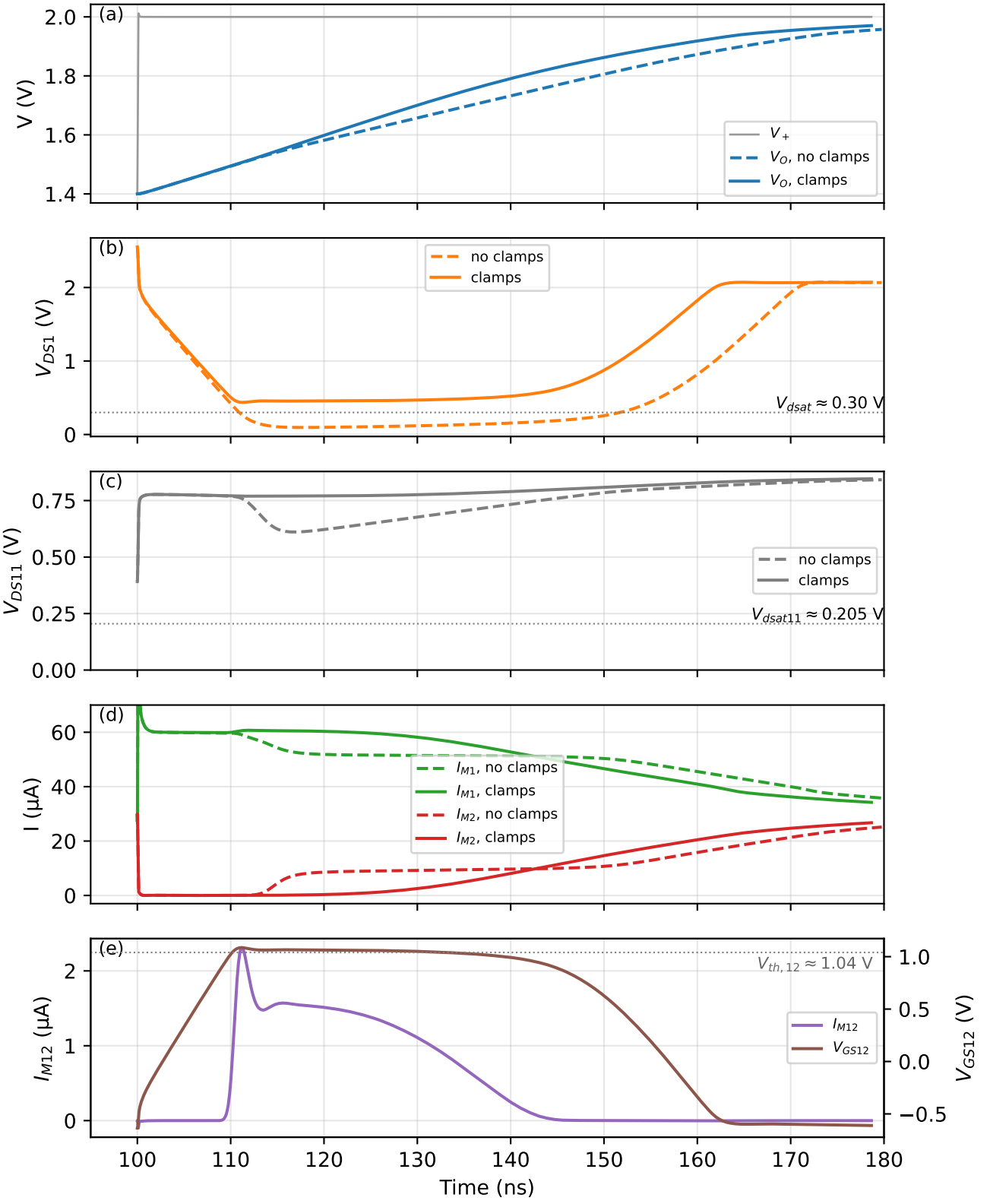
(a)
V (V)
2.0
1.8
1.6
1.4
V_+
V_O, no clamps
V_O, clamps
(b)
V_{DS1} (V)
2
1
0
no clamps
clamps
$V_{dsat} \approx 0.30$ V
(c)
V_{DS11} (V)
0.75
0.50
0.25
0.00
no clamps
clamps
$V_{dsat11} \approx 0.205$ V
(d)
I (µA)
60
40
20
0
I_{M1}, no clamps
I_{M1}, clamps
I_{M2}, no clamps
I_{M2}, clamps
(e)
I_{M12} (µA)
2
1
0
$V_{th,12} \approx 1.04$ V
I_{M12}
V_{GS12}
V_{GS12} (V)
1.0
0.5
0.0
−0.5
100
110
120
130
140
150
160
170
180
Time (ns)